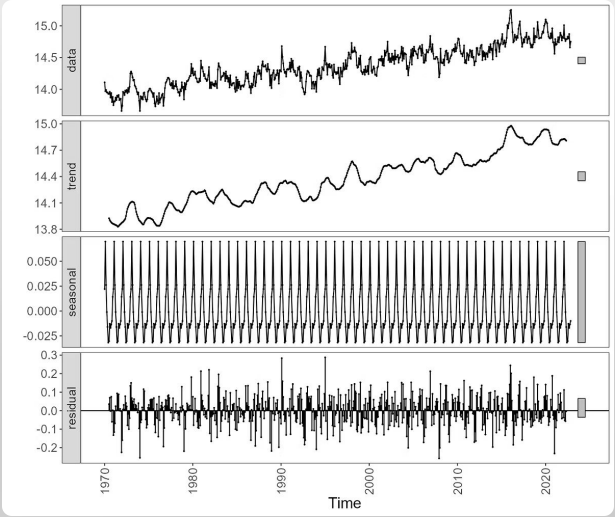
Time Series Decomposition: Trend, Seasonality, and Residuals Explained

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Time Series Decomposition



The Core Idea

Decomposition separates X_t into interpretable components. Each component represents a distinct aspect of temporal behavior. The combination form depends on how amplitudes behave over time.

$$X_t = T_t + S_t + \varepsilon_t$$

- T_t — **trend**: long-term behavior
- S_t — **seasonality**: regular cyclic patterns with period s
- ε_t — **residual**: variations not explained by the other components

Additive model: constant amplitudes over time.
Multiplicative model: amplitudes proportional to the level — equivalent to additive after $\log(X_t)$.

> The residual ε_t captures what was not explained by the structure — it is where events and anomalies may manifest.

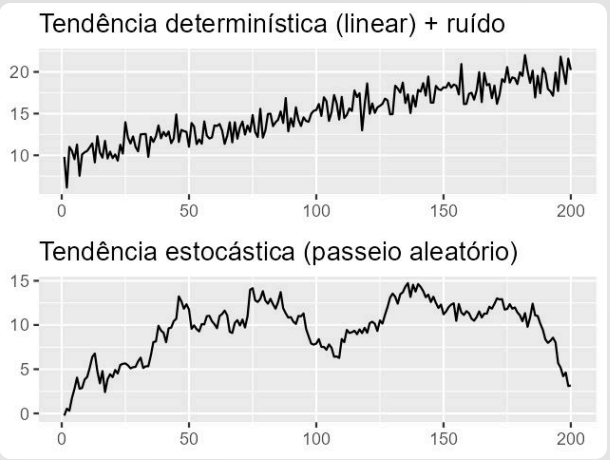
INPUT: x_t (t = 1,...,T), s (seasonal period), model ∈ {additive, multiplicative}
PRE-CONDITION: s known; model chosen based on the series plot

STEP 1 — Estimate trend
T̂_t ← moving average of x_t with window s

STEP 2 — Estimate seasonality
r_t ← x_t − T̂_t (additive) | x_t / T̂_t (multiplicative)
Ŝ_p ← mean of r_t with (t mod s) = p, p = 1,...,s
Normalize Ŝ: zero sum (additive) or unit product (multiplicative)

STEP 3 — Obtain residual
ε_t ← x_t − T̂_t − Ŝ_t (additive) | x_t / (T̂_t · Ŝ_t) (multiplicative)

OUTPUT: T̂_t, Ŝ_t, ε_t for each t
POST-CONDITION: ε_t with no systematic pattern and approximately stationary



Trend Models

Trend represents the systematic long-term variation in a time series — the expected value $T_t = \mathbb{E}[X_t]$ that changes over time in non-stationary series. Identifying it is essential to separate permanent movements from transient fluctuations.

Linear Trend

Constant growth or decline over time

$$T_t = \alpha + \beta t$$

Polynomial Trend

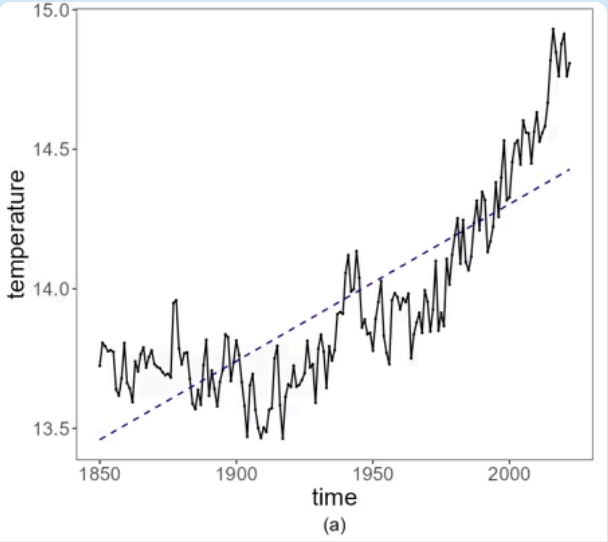
Non-linear variations described by polynomial functions

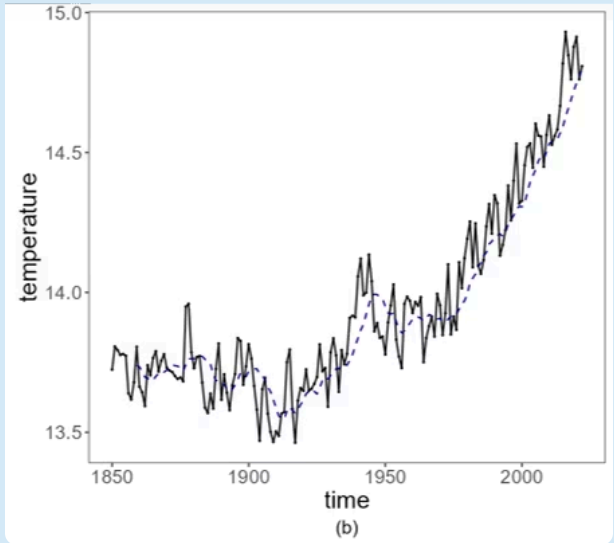
Stochastic Trend

Cumulative random evolution

$$T_t = T_{t-1} + u_t$$

Trend models can be deterministic, when defined by explicit functions of time, or stochastic, when they evolve randomly. This distinction is fundamental for defining appropriate estimation and transformation methods.





Linear Regression and Moving Average

Linear regression

- Assumes a simple global form.
- Good for approximately linear trends.
- Facilitates interpretation.

Moving average

- Uses local neighborhood.
- Smooths short-term noise.
- Depends on window size.



Seasonality: Periodic Patterns

Seasonality corresponds to patterns that repeat regularly over time, such as monthly, quarterly, or weekly variations. Unlike trend, it does not represent a permanent structural change.

The formal periodicity condition is expressed by:

$$S_{t+s} = S_t$$

Where s represents the seasonal period. This property distinguishes seasonality from other temporal components.

Common Seasonal Periods

Monthly Data

Annual seasonality in series with monthly observations — period s = 12

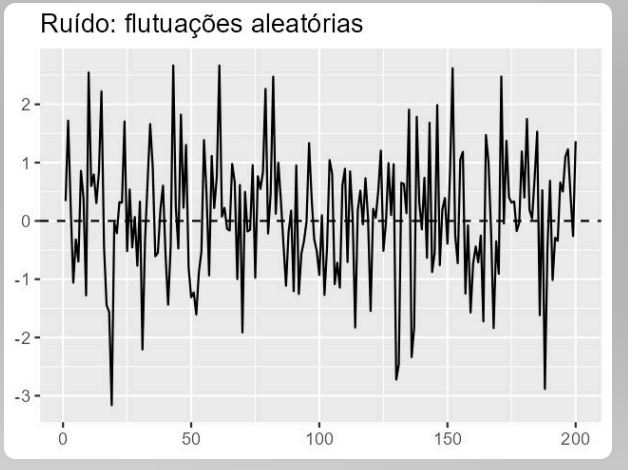
Quarterly Data

Annual pattern in quarterly series — period s = 4

Weekly Data

Weekly pattern in daily data — period s = 7

The seasonal period s depends on the observation frequency of the series. These values formalize periodicity and allow seasonality to be incorporated into statistical models precisely.



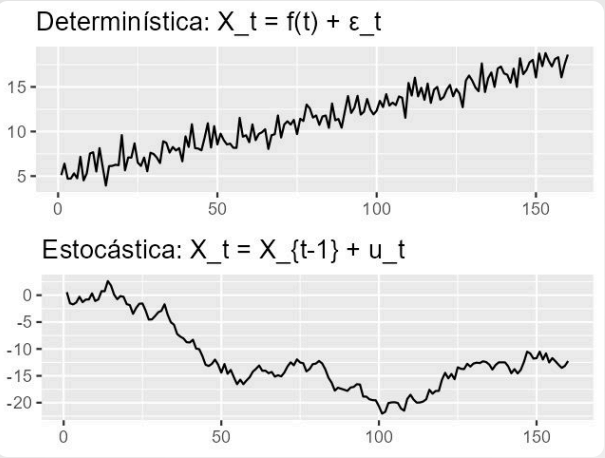
Random Component: The Noise

The random component represents unpredictable fluctuations that cannot be explained by trend, seasonality, or cycles. It is the noise inherent to the observed process.

In classical models, noise is assumed to have zero mean and constant variance:

$$\mathbb{E}[\varepsilon_t] = 0, \quad \mathrm{Var}(\varepsilon_t) = \sigma^2$$

These assumptions simplify the analysis, but can be relaxed in more sophisticated models that capture heteroscedasticity and temporal dependence in the noise.



Deterministic vs. Stochastic Trend

Trend can be deterministic — following a predictable trajectory — or stochastic — evolving randomly and cumulatively. This distinction defines how it should be removed.

Deterministic

Predictable and systematic trajectory

The series follows a known function of time — linear, quadratic, or exponential. The future is predictable from past patterns.

$$X_t = f(t) + \varepsilon_t$$

Removal by regression: fit $f(t)$ and subtract from the series.

Stochastic

Random and cumulative drift

The series accumulates random shocks over time, without returning to a fixed level. The future is unpredictable even knowing the past.

$$X_t = X_{t-1} + u_t$$

Removal by differencing: compute $\Delta X_t = X_t - X_{t-1}$.

Identifying the type of trend is the first step before any modeling — the wrong method can distort the entire analysis.

Trend Removal

The Core Idea

The removal method depends on the type of trend. Applying the wrong method distorts the entire subsequent analysis.

Deterministic: fit $f(t)$ by regression → $Y_t = X_t - \hat{T}_t$

Stochastic: differencing → $Y_t = X_t - X_{t-1}$

Verify stationarity after removal (e.g., ADF test).

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INPUT: x_t (t = 1,...,T), type ∈ {deterministic, stochastic}
PRE-CONDITION: trend type identified beforehand

STEP 1 — Removal according to type
  IF type = deterministic:
    T̂_t ← regression of x_t on t  (linear, polynomial, etc.)
    Y_t  ← x_t − T̂_t

  IF type = stochastic:
    Y_t  ← x_t − x_{t−1}   (t = 2,...,T)

STEP 2 — Verification
  Test stationarity of Y_t  (e.g., ADF test)
  IF not stationary: repeat differencing

OUTPUT: Y_t (detrended series)
POST-CONDITION: Y_t approximately stationary in mean
```

Seasonality Removal

The Core Idea

Deseasonalization removes the periodic pattern S_t from the series, isolating the dynamics not explained by seasonality.

Additive: $Y_t = X_t - \hat{S}_t$

Multiplicative: $Y_t = X_t / \hat{S}_t$

Y_t is the deseasonalized series — the basis for modeling trend and noise.

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INPUT: x_t (t = 1,...,T), s (seasonal period), model ∈ {additive, multiplicative}
PRE-CONDITION: s known; trend removed or controlled

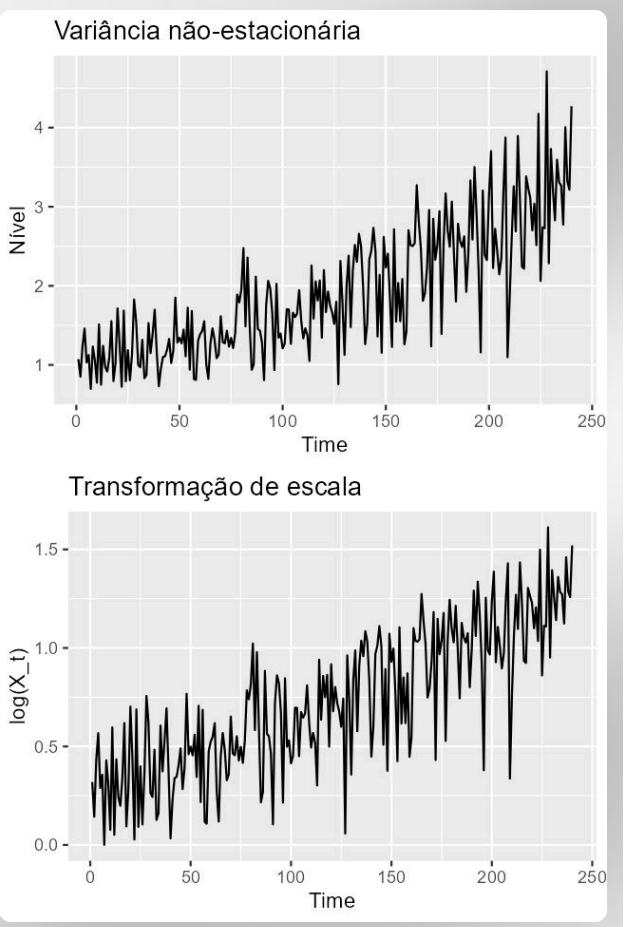
STEP 1 — Identify seasonal period s
  (monthly: s=12 | quarterly: s=4 | daily: s=7)

STEP 2 — Estimate seasonal component
  FOR p = 1,...,s:
    Ŝ_p ← mean of x_t with (t mod s) = p
  Normalize: zero sum (additive) | unit product (multiplicative)

STEP 3 — Remove seasonality
  Y_t ← x_t − Ŝ_{(t mod s)}   (additive)
  Y_t ← x_t / Ŝ_{(t mod s)}   (multiplicative)

STEP 4 — Verify residual
  Y_t should not exhibit significant seasonal autocorrelation

OUTPUT: Y_t (deseasonalized series)
POST-CONDITION: absence of periodic pattern in Y_t
```



Variance Stabilization

Some series exhibit conditional variance $\sigma_t^2 = \mathrm{Var}(X_t \mid \mathcal{F}_{t-1})$ that changes over time — a phenomenon called **heteroscedasticity**. Transformations aim to make this variance approximately constant before modeling or forecasting.

$$\mathrm{Var}(Y_t \mid \mathcal{F}_{t-1}) \approx \text{constant}$$

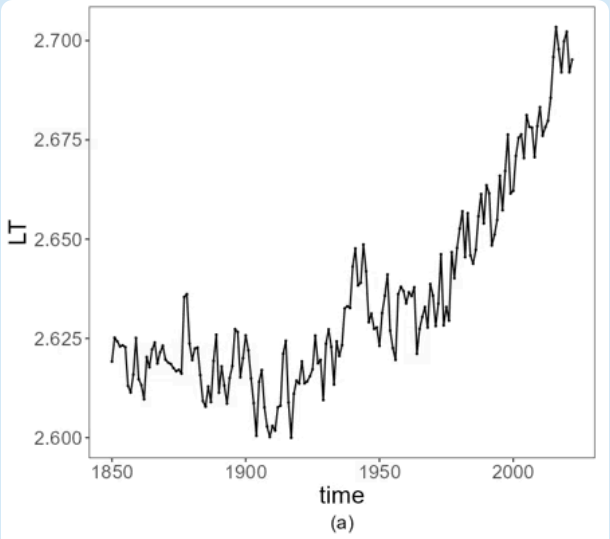
Logarithmic Transformation

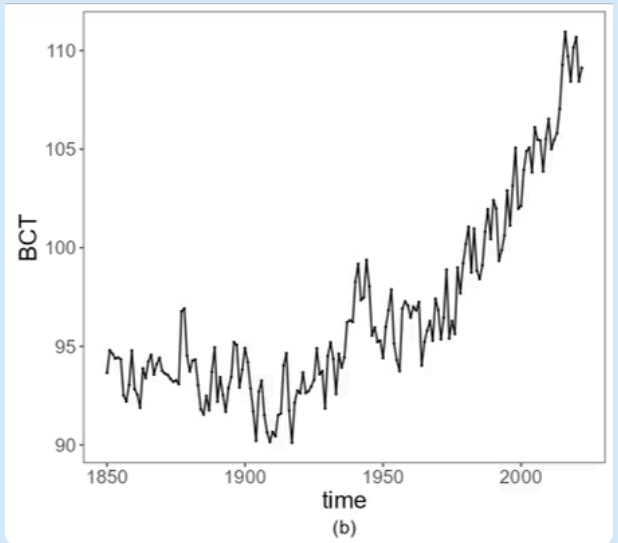
When variance grows proportionally to the level: $Y_t = \log(X_t)$. Stabilizes series with growing amplitudes.

Box-Cox Transformation

Parametric family: $Y_t = (X_t^\lambda - 1)/\lambda$. For $\lambda = 0$ it equals the logarithm; $\lambda = 1$ keeps the original series.

⚠ Unstable variance directly affects **prediction intervals**: confidence bands become poorly calibrated. Furthermore, periods of high volatility can be mistaken for real events, generating **false positives in detection**. Stabilizing variance is a prerequisite for reliable analysis.





Log Transform and Box-Cox

Log transform

- Reduces the amplitude of high values.
- Useful for multiplicative growth.
- Requires positive values.

Box-Cox

- Parameterized family of transformations.
- Chooses λ to stabilize variance.
- More flexible than simple log.

Decomposition and Analysis

Each decomposition method generates a different representation of the time series. The choice of method is not neutral: it defines what will be seen as a pattern, what will remain in the residual, and therefore what can be detected as a deviation.

The same point can be noise in one decomposition and an anomaly in another. The residual is not raw data — it is the result of an analytical decision.

> There is no neutral residual. Decomposition defines what is considered "normal". Choosing the method is choosing what will be seen.

Method defines the normal

The extracted structure (T_t, S_t, C_t) determines what is expected. The residual ε_t is what remains of that expectation.

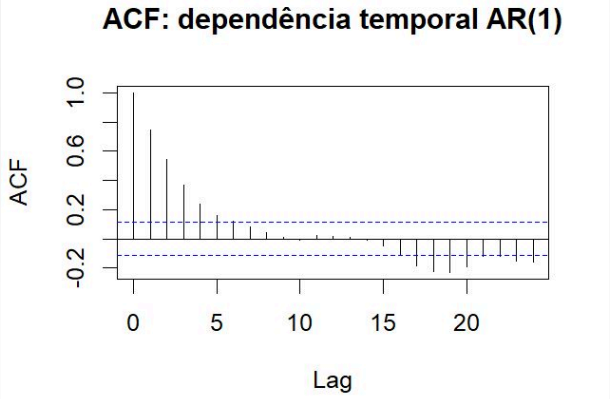
Residual as the analysis space

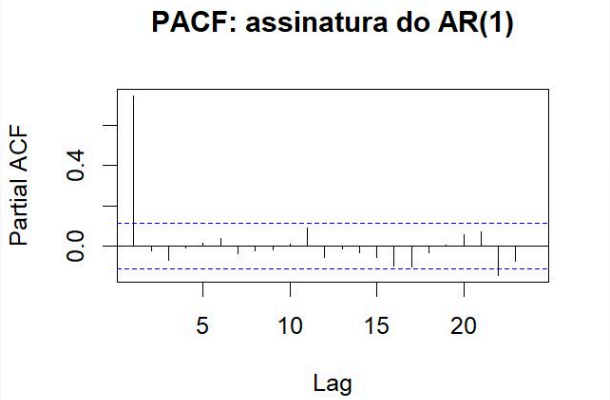
It is in ε_t that events, anomalies, and deviations may manifest. The quality of the decomposition determines the quality of the analysis.

Bridge to models

Understanding the structure of the series is the prerequisite for modeling, forecasting, and detection. Pipelines, features, and models are covered in the next lecture.

Autocovariance and Autocorrelation





Time series have memory: nearby observations tend to be more similar than distant ones. Autocovariance and autocorrelation quantify this memory.

Autocovariance γ(h)

Linear association between X_t and X_{t-h}.

$$\gamma(h) = \mathrm{Cov}(X_t,\, X_{t-h})$$

Autocorrelation ρ(h)

Standardized version of γ(h), always in [−1, 1].

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)}$$

Short-Range Dependence

ρ(h) decays quickly to zero — limited memory.

Ex.: white noise.

Long-Range Dependence

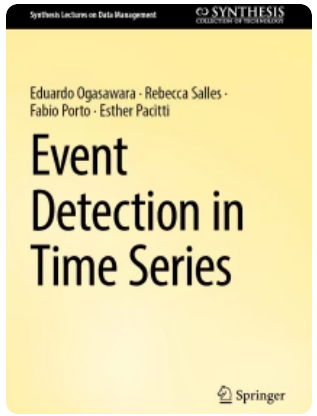
ρ(h) decays slowly (hyperbolically) — past events still influence the present.

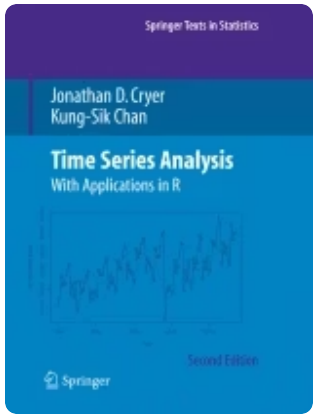
Note: $\rho(h) \sim h^{-\alpha}, 0 < \alpha < 1.$

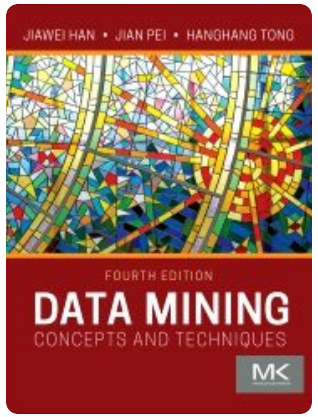
Ex.: climate and financial series.

The autocorrelation function (ACF) is the main diagnostic tool for identifying the type of dependence and guiding model selection.

References







Event Detection in Time Series

Ogasawara, E.; Salles, R.; Porto, F.; Pacitti, E. Springer Nature Switzerland, 2025.

Time Series Analysis: With Applications in R

Cryer, J. D.; Chan, K.-S. Springer, 2008.

Data Mining: Concepts and Techniques

Han, J.; Pei, J.; Tong, H. Morgan Kaufmann, 4ª ed., 2022.